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AI-Driven Slip Model Integration in Embedded Edge Devices for Accurate Multiphase Flow Measurement

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Abstract

This work presents a machine learning-based slip model designed for real-time integration into multiphase flow meters operating on embedded edge devices. Unlike traditional slip models constrained by fixed correlations, the proposed approach leverages physics-informed features and is trained on diverse flow loop datasets. The resulting model, deployed as a lightweight Java Archive (JAR) and interfaced with C++ via Java Native Interface (JNI), achieves sub-15 millisecond inference latency within limited hardware resources. Field and laboratory trials confirm the predictive accuracy and operational stability of this approach across a wide range of flow regimes, demonstrating its potential to enhance the fidelity and responsiveness of embedded multiphase metering systems.

Introduction

Accurate multiphase flow measurement (MPFM) is essential for production optimization, reservoir management, and allocation reporting, particularly in unconventional assets characterized by wide fluctuations in flow regimes. A longstanding challenge in multiphase metering is the estimation of slip ratio (S), defined as the velocity ratio between gas and liquid phases (1):

$$S = \frac{Effective\ \ Gas\ \ Phase\ \ Velocity}{Effective\ \ Liquid\ \ Phase\ \ Velocity} \tag{1}$$

Slip directly impacts phase holdup and flow rate allocation, yet is inherently nonlinear and sensitive to fluid properties, pipe geometry, and flow regime transitions.

Existing slip models are largely based on empirical correlations derived from limited datasets or constrained flow loop experiments. Classical approaches such as those proposed by Zuber and Findlay (1965), Chisholm (1973), and Smith (1975) provide convenient analytical expressions but rely on simplifying assumptions, including uniform flow, fixed orientation, or consistent phase distribution. These models often break down under field conditions, particularly at high gas volume fractions (GVF > 90%) or in transitional and mist flow regimes where phase interactions become complex (Hasan and Kabir, 2002;

Ansari et al., 1994). Moreover, their rigid formulations do not adapt to varying pipe diameters, inclination, or evolving well conditions.

In recent years, data-driven methods have emerged as promising alternatives, offering greater flexibility in modeling nonlinear, high-dimensional systems. However, purely statistical machine learning (ML) models can be prone to overfitting, lack physical interpretability, and require extensive retraining when operational conditions change (Scholkopf and Smola, 2002). This motivates the development of a hybrid approach that leverages domain knowledge for input selection while maintaining the flexibility of machine learning techniques.

This study introduces a machine learning-based slip prediction model that is guided by physics-informed feature engineering. Key features include gas-liquid density ratio ($\rho g/\rho l$), Lockhart-Martinelli parameter (XLM), and gas and liquid phase Froude numbers (Frg) parameters rooted in fluid mechanics and dimensionless analysis. These features enable the model to generalize across a wide operational envelope without relying on flow regime maps or fixed model coefficients.

The model is trained on extensive external flow loop datasets and validated under both controlled and field conditions. Unlike traditional models, it is designed for real-time execution on embedded edge devices with strict memory and latency constraints. This deployment architecture was achieved using Java Native Interface (*JNI*) to integrate the model, compiled as a JAR file, into the existing C++ flow computation framework used in the meter.

The resulting system delivers sub-15 millisecond inference latency with high prediction accuracy, enabling real-time updates of phase slip without requiring recalibration or manual tuning. This capability addresses key industry pain points and advances the state of embedded multiphase metering technology by coupling rigorous feature selection with low-latency deployment in resource-constrained environments.

Methodology

The model development followed a structured, physics-informed workflow spanning from data preparation to real-time deployment on embedded systems.

Data Preparation and Feature Engineering

Multiphase flow loop data was aggregated from internal and external testing campaigns. Inputs included structured Excel logs, CSV exports, and flat-file records, all consolidated into an SQLite database. The final dataset covered a wide range of conditions, including gas volume fractions (GVF) from 10% to 99.9%, water liquid ratios (WLR) from 0% to 100%, and pipe diameters ranging from 2 to 8 inches.

To ensure high-quality training data, raw records were filtered for duplicate timestamps, unstable test points, and known sensor faults. Stabilization logic was applied to exclude transient startup or shutdown periods, retaining only steady-state operating conditions. Data was further segmented by pipe diameter, fluid configuration, and GVF range to enable stratified sampling during training and testing.

Data preprocessing and structuring were performed using Jupyter in a Codespaces environment. Feature engineering emphasized dimensionless groups derived from established multiphase flow principles. The slip ratio was modeled as a function of key physically relevant parameters:

$$S = f(XLM, \quad \rho g / \rho l, \quad Frg, \ldots) \tag{2}$$

Where $f(.)$ represents a data-driven mapping trained to capture the relationship between flow structure and phase velocities. Input variables were selected to reflect momentum balance, inertial scaling, and interfacial interactions, without relying on discrete flow regime classification. Only a representative subset of these features is shown in Equation (2) for clarity.

Statistical screening using Pearson correlation, mutual information, and normalized ACE scores was applied to assess feature relevance, reduce redundancy, and prevent target leakage, consistent with established methodologies in feature selection (Guyon and Elisseeff, 2003).

Model Selection and Training

Over 50 supervised learning algorithms were evaluated using AutoML pipelines. These included linear regressors, ensemble models, and kernel-based methods. Model selection prioritized cross-validation root mean squared error (RMSE), generalization error, and computational footprint.

The final model was a Support Vector Regressor (SVR) with a radial basis function (RBF) kernel approximated via the Nystroem method, which enables scalable approximation of nonlinear behavior (Schölkopf and Smola, 2002). Final hyperparameter tuning included a constrained grid search over kernel width (gamma), regularization penalty (C), and number of basis components. The model was explicitly limited in complexity to ensure generalization across pipe diameters and fluid types.

A 5-fold cross-validation scheme was used, stratified by pipe size and GVF band. This approach ensured performance consistency across underrepresented conditions and avoided overfitting to dominant operational segments. Similar ML approaches have demonstrated value in flow rate allocation workflows for virtual metering in the field (Luo et al., 2015).

Model Deployment

The final model could be deployed either as a Java JAR file or as part of a docker container that serves inference results via a REST API. Due to integration requirements detailed in section 2.4, the selected model was exported in Java Archive (JAR) format and initially tested using MATLAB as a prototype environment. MATLAB served exclusively for early-stage validation and performance profiling. To ensure consistency with the field system, the same core C++ codebase used in the embedded device was also called from MATLAB during development.

This dual-platform approach enabled numerical consistency checks between legacy slip correlation logic and the new ML model. Output matching across platforms confirmed that the JAR-based implementation adhered to the meter's real-time requirements and data structure expectations.

Embedded Integration and Resource Optimization

For real-time deployment, the model needed to be integrated into the MPFM Flow Computer embedded software. This software is deployed onto a ruggedized edge controller running Windows Embedded OS with a 1.33 GHz processor, 2GB of Ram, and 30 GB of onboard storage. Within this constrained environment, the flow computation engine operates with <50 milliseconds per cycle, requiring that the model return an inference result within 15 milliseconds. Additionally, the model code needed to be compatible with the existing core C++ codebase.

As previously mentioned, the model could be deployed either as a Java JAR file or as part of a docker container that serves inference predictions via a REST API. While C++ code can interact with REST APIs, the docker container approach requires more resources for the container to run (particularly memory and storage). These requirements were too large for the small amount of resources that the edge controller had available. Furthermore, there were concerns that the REST API could return an inference result fast enough to meet the response time requirements.

The Java JAR file export of the model on the other hand, requires less resources to run as it only needs the JAR file and a Java Runtime Environment to be able to execute. The model JAR file is also designed for lower-latency inference due to the more direct access to the model logic. This method, however, does require more software development. Ultimately, integrating the model in as small a package as possible was the more paramount concern, so the JAR file method was used.

The remaining piece of the deployment integration strategy was then how to bridge the language gap and have C++ communicate with Java. To achieve this, the Java Native Interface (JNI) was chosen. Typically, this foreign function interface is used by developers to allow Java to call and interact with libraries or applications that are coded in other languages, such as C or C++. But it does support instantiating connections from languages such as C++ to Java and was the most performant method found during research for this integration. An additional dedicated Java abstraction layer was developed to decouple the C++ interface from the model code. This allows for easy updates to the model without modifying the existing firmware or recalibration logic.

Memory profiling was conducted across heap size allocations from 20 MB to 512 MB. Lower values led to intermittent latency spikes due to increased Java garbage collection, while higher limits risked resource contention with other onboard processes. A final cap of 120 MB was selected.

To further minimize memory pressure while preserving throughput, the following runtime strategies were implemented:

- **Single-instance model loading**: The ML model was instantiated once during system startup and retained in memory throughout runtime. This avoided repeated loading penalties and ensured thread-safe reuse of the model object across flow computation cycles.
- **Explicit memory release after each prediction**: Temporary data structures created during inference were immediately de-referenced after use, reducing the persistence of short-lived objects and minimizing memory fragmentation within the Java Virtual Machine (JVM).
- **Buffer reuse to suppress unnecessary garbage collection**: Input and output arrays were preallocated and reused across calls. This prevented frequent heap allocations, significantly reducing garbage collection frequency and associated latency spikes.

These measures ensured deterministic low-latency execution and stability for continuous real-time metering under dynamic flow conditions (Choi et al., 2021).

FlowLoop Trial

To evaluate the embedded Slip_ML model under controlled test conditions, the MPFM system was deployed in an independent external flow loop. The test matrix spanned a wide operational envelope, including gas volume fractions up to 99.9%, variable water cuts, and multiphase flow regimes ranging from slugging to annular conditions. The trial verified the fidelity of integration and computational stability under dynamic but repeatable input conditions, simulating edge deployment scenarios.

Field Trial

Following laboratory validation, the upgraded MPFM was deployed in an unconventional gas field for blind testing against a calibrated test separator. The site presented high-GVF conditions (above 80%) and water cuts up to 70%, typical of complex production environments. This deployment allowed the embedded model to be exercised under field conditions not reproducible in a flow loop, including varying choke sizes, wellhead pressures, and production transients, thereby completing the end-to-end validation cycle under real-world constraints.

Results and Discussion

The performance evaluation of the developed slip model focused on both statistical accuracy and operational feasibility. This section presents detailed metrics, benchmarking outcomes, and interpretability analysis to assess the model's ability to generalize across flow regimes while meeting the latency and resource constraints required for real-time deployment in multiphase flow meters.

Model Metrics and Performance Analysis

The final Support Vector Regressor (SVR) with a radial basis function (RBF) kernel, approximated via the Nystroem method, was benchmarked against over 50 candidate algorithms. These included ensemble methods (e.g., XGBoost, Random Forest), linear regressors (e.g., Ridge, Lasso), and gradient-boosted tree models. Each model was evaluated using key performance metrics: root mean squared error (RMSE), mean absolute error (MAE), mean absolute percentage error (MAPE), and normalized Gini coefficient (Gini Norm) across holdout and cross-validation sets.

The selected Nystroem Kernel SVR demonstrated the best overall trade-off between predictive accuracy, generalization stability, and real-time deployment suitability. Specifically, the model achieved:

- RMSE (Holdout): 0.2204
- MAE (Holdout): 0.2520
- MAPE (Holdout): 6.74%
- Gini Norm (Holdout): 0.9497
- Average Inference Latency: <15 ms per prediction

Cross-validation using 5-folds confirmed consistent performance across stratified data segments, with $R^2 = 0.8635$ and average RMSE ≈ 0.252. A standard deviation of ≈0.12 across folds demonstrated strong model stability and generalization. A comparative summary of leading candidates is provided in Table 1, illustrating that although XGBoost achieved slightly higher Gini Norm, it incurred significantly higher latency, which disqualified it from real-time deployment. Nystroem Kernel SVR's superior balance of accuracy and edge feasibility.

Table 1—Model Benchmarking Results on Holdout Dataset

Model	RMSE	MAE	MAPE	Gini Norm
Nystroem Kernel SVM Regressor	0.2204	0.252	6.7446	0.9497
XGBoost Regressor	0.222	0.2627	6.5539	0.952
Nystroem Kernel SVM Regressor (different hyperparameters)	0.229	0.2495	6.8697	0.9492
Random Forest Regressor	0.2426	0.3105	7.0279	0.9446
Linear Regression	0.4439	0.4912	7.0279	0.8201

Beyond summary metrics, the SVR's tracking behavior was assessed over 60 validation bins spanning the entire operational envelope. The SVR closely followed the actual slip trend, especially in nonlinear high-GVF regimes (e.g., bins 1–5 and 55–60), where multiphase interactions become most pronounced. In contrast, linear regression models consistently underestimated slip in these transitional regimes, highlighting their limitations in capturing multiphase complexity as illustrated in Figure 1.

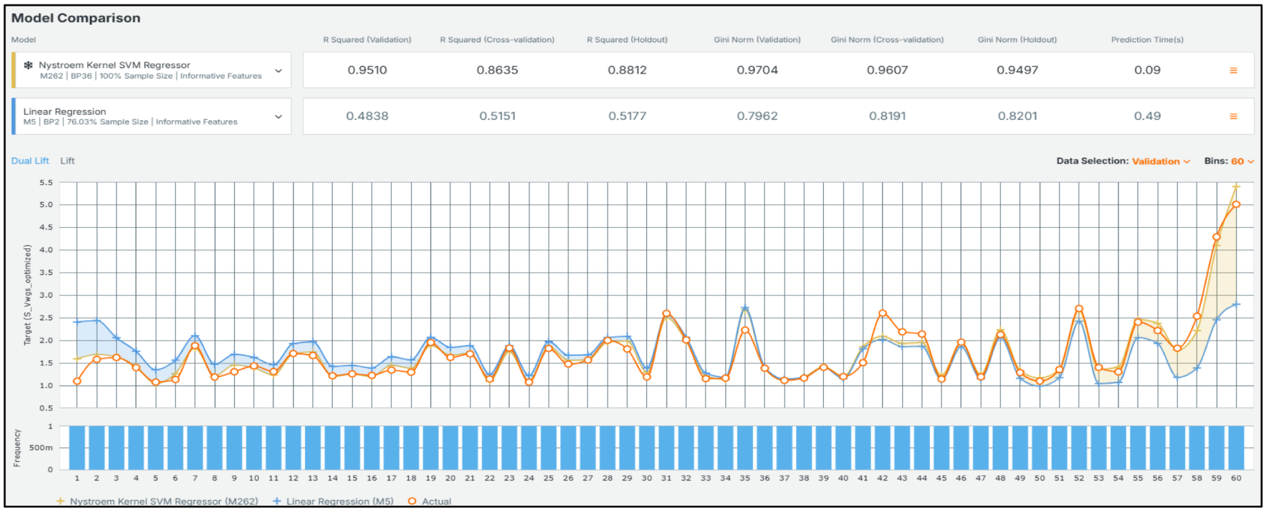

Figure 1—Slip Prediction Accuracy Across Validation Bins: Nystroem Kernel SVR vs. Linear Regression

Latency considerations were equally critical for embedded deployment. When plotted against validation RMSE, the SVR model maintained the lowest error while achieving inference latency well below 15 ms per prediction. Competing models such as XGBoost and Random Forest delivered comparable RMSE values but at a latency cost of 400 to 500 ms orders of magnitude higher and incompatible with edge computing requirements as illustrated in Figure 2.

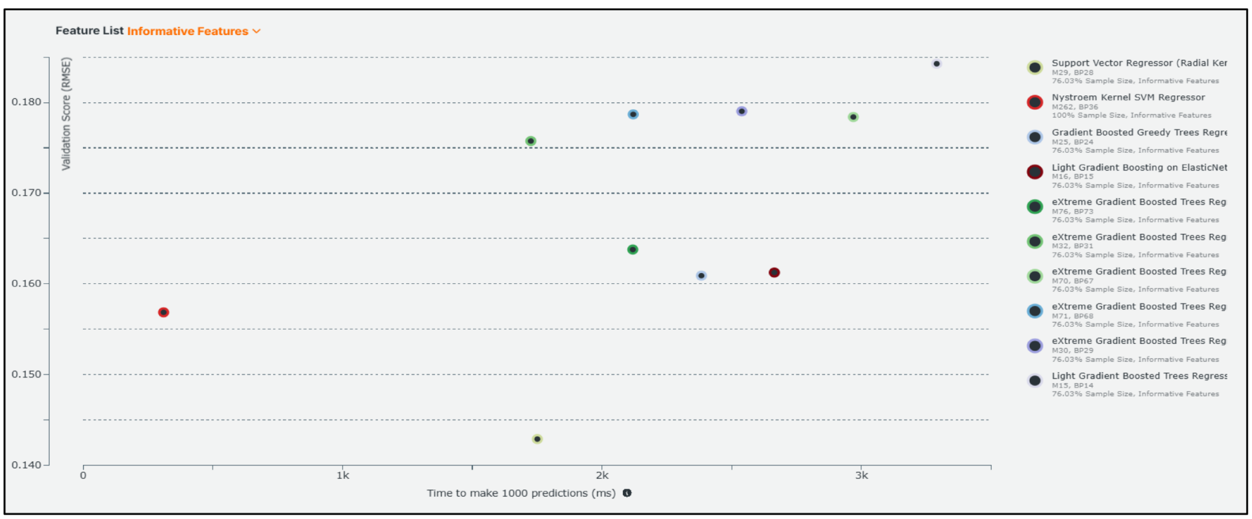

Figure 2—Latency vs. Validation RMSE Trade-Off for Candidate Models

To ensure physical plausibility and interpretability, partial dependence analysis was performed on the final trained SVR model. This method isolates the marginal effect of each input variable by holding all others constant, enabling clear attribution of feature influence on predicted slip values. The analysis revealed that:

- Frg (gas-phase Froude number) exhibited a strong monotonic increase, reflecting the physics of rising slip ratios with higher gas momentum, indicative of intensified interfacial shear and phase separation.
- Xlm (Lockhart-Martinelli parameter) showed a positively nonlinear trend, consistent with greater slip under increased liquid loading.

These relationships align with established multiphase flow behavior, indicating that the model captures physically meaningful patterns. While the SVR is not explicitly physics-based, its learned behavior mirrors flow regime theory, striking a balance between data-driven flexibility and physical interpretability.

Embedded Deployment Performance

Following integration via the Java Native Interface (JNI), the slip ML model's runtime behavior was assessed under constrained embedded conditions representative of field deployments. The evaluation platform consisted of a 32-bit Windows 7 Embedded system with 2 GB of RAM, reflecting the target hardware used in edge-based multiphase flow metering applications.

When running the model with default settings in Java, which included no user-defined limit for heap memory usage, erratic and high memory consumption was observed. Because this model needed to run on a limited shared resource, this was not a viable configuration. Tests were then performed to determine the boundaries of how little memory the model could use while still consistently meeting the 15 millisecond inference requirement.

In tests with heap memory limits capped below 100 MB, prediction instability could be occasionally seen. This manifested as either longer times for an inference to be made or as an execution failure of the Java code itself. Investigation revealed that this was typically from Java garbage collection running more frequently (due to less memory available to work with), while execution failures occurred if garbage collection could not free up enough memory. Conversely, heap memory sizes above 150 MB provided more consistent inference latency times but posed a risk to overall system health by competing for memory resources with co-resident services on the embedded unit.

A balanced configuration with a capped heap memory limit of 120 MB was ultimately decided upon. In tests with this limit, the model consistently met requirements for inference time to be below 15 milliseconds per call across all test scenarios.

The deployed model sustained continuous operation over 100,000 iterations without degradation in performance or stability. It performed reliably in both batch-inference routines and real-time, single-prediction modes. This validated the model's suitability for field deployment on constrained platforms, enabling standalone real-time flow diagnostics without reliance on cloud infrastructure or high-performance computing resources.

External flow-loop validation

To quantitatively evaluate performance under controlled test conditions, error statistics were computed by comparing model-predicted and gravimetrically measured gas and liquid rates using data from an independent external flow loop. Two model variants were assessed: the Slip_ML model trained with domain-specific data and a generic non-custom model representative of correlation-based, an off-the-shelf implementation.

The Slip_ML model consistently demonstrated improved accuracy in both gas and liquid predictions. As shown in Table 2, the gas rate MAE was reduced from 5.14% to 2.93% (a 43% improvement), and the mean error was minimized from 2.19% to −0.75%, indicating reduced bias. However, the most notable performance gain was observed in liquid rate prediction, which is often more sensitive to flow regime transitions. The MAE dropped from 13.77% to 6.93%, nearly a 50% improvement, while the mean error improved from −11.15% to −7.19%, reflecting better tracking of the liquid phase in high-GVF and transitional conditions.

Table 2—Summary of Error Statistics from Independent Flow Loop Evaluation

Error Metric	Gas Rate (Non-Custom)	Gas Rate (Slip_ML)	Liquid Rate (Non-Custom)	Liquid Rate (Slip_ML)
Mean Error [%]	2.19	−0.75	−11.15	−7.19
Mean Absolute Error (MAE) [%]	5.14	2.93	13.77	6.93
Standard Deviation [%]	6.02	4.31	13.35	8.62

Importantly, the Slip_ML model maintained a tighter error distribution. The standard deviation in gas rate error decreased from 6.02% to 4.31%, and for liquid rate, from 13.35% to 8.62%. The interquartile range (IQR) also narrowed from 4.29% to 3.34% for gas and from 10.82% to 7.86% for liquid, confirming enhanced prediction stability across a range of flow regimes, including transitional and liquid-dominated conditions.

This comparison highlights that models not tailored to the operating envelope may underperform when applied to conditions beyond their original development scope. While not reflective of any specific deployed implementation, the non-custom model serves to underscore the importance of adaptation. The results validate the Slip_ML model's ability to generalize beyond its training data while preserving physical realism and deployment feasibility.

To further illustrate this improvement, Figure 3 presents the liquid rate error plotted against reference GVF for both models. The Slip_ML model exhibits tightly clustered error values. In contrast, the non-custom model shows significantly higher variance, with scattered over- and under-predictions especially pronounced between 90% and 97% GVF. This reinforces the importance of model adaptation to the operating envelope and underscores the limitations of applying static correlations beyond their original design conditions.

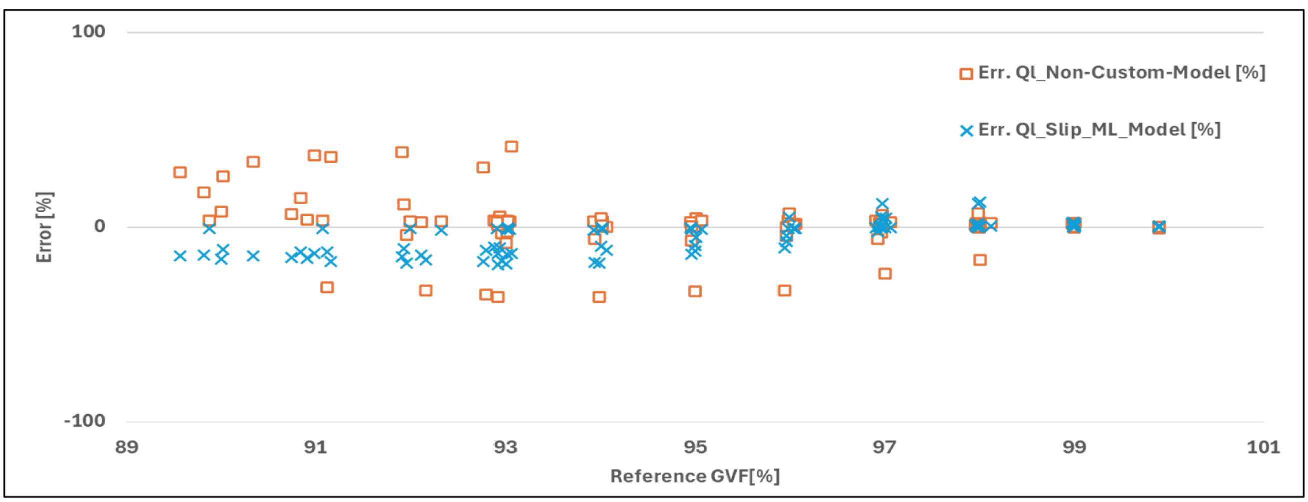

Figure 3—Liquid Rate Error vs. GVF: Slip_ML Model vs. Non-Custom Model

Field Validation in Unconventional Gas Field

The upgraded multiphase flow meter (MPFM), integrated with the Slip_ML model, was field-tested in an unconventional gas field to evaluate its performance under real production conditions. The test was conducted as a blind trial against a calibrated test separator, ensuring unbiased performance assessment.

The field environment featured a wide operational envelope. The gas volume fraction (GVF) spanned 80–99%, while the water cut ranged from 40 to 70%, reflecting challenging multiphase flow regimes. Gas rates varied from 1 to 5 MMSCFD, and liquid rates ranged from 500 to 4,000 BPD. These variations provided a robust dataset across slugging, transitional, and high-GVF conditions.

The field validation was conducted under various choke sizes and pressures to ensure that the test results were accurate and reflective of real operating conditions, which are often underrepresented in lab-scale trials. By varying the choke sizes during the test, the research team could evaluate the MPFM's performance under different flow regimes, thereby broadening the understanding of how the meter responds to altering production scenarios.

Importantly, the meter's inference cycle time stayed below 15 ms throughout the field test, meeting real-time diagnostic requirements. No instability or memory overload was observed, confirming that embedded deployment constraints identified during lab testing remained valid in the field.

Results showed that the Slip_ML model enabled the MPFM to meet customer performance specifications in all tested scenarios, with gas rate error consistently within ±5% and liquid rate error within ±10%. This confirmed both the model's predictive accuracy and its reliability under dynamic field conditions without requiring field-specific recalibration.

Conclusion

This study demonstrates the successful development, deployment, and validation of a physics-informed machine learning slip model (Slip_ML) designed for real-time integration within multiphase flow meters. The model architecture, built on Support Vector Regression with physically motivated features, outperformed traditional and generic approaches in both controlled flow loop and field scenarios. It achieved superior predictive accuracy, reducing mean absolute error by up to 50%, and maintained tight error bounds across a wide range of flow regimes.

Notably, the model was embedded within an edge device and optimized for deterministic performance, achieving an inference latency of under 15 ms under constrained memory conditions. Its deployment into a fielded MPFM unit in an unconventional gas environment confirmed both operational feasibility and generalization beyond training data, with gas and liquid flow rate errors remaining within ±5% and ±10%, respectively, meeting stringent customer specifications without the need for field recalibration.

By closing the loop from feature engineering to embedded deployment, this work advances the practical application of intelligent models in harsh production environments, offering a scalable path forward for high-fidelity, low-latency multiphase metering.

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